



Principle of Economics

Chapter Four

Theory of Production and Cost

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3.1. Introduction

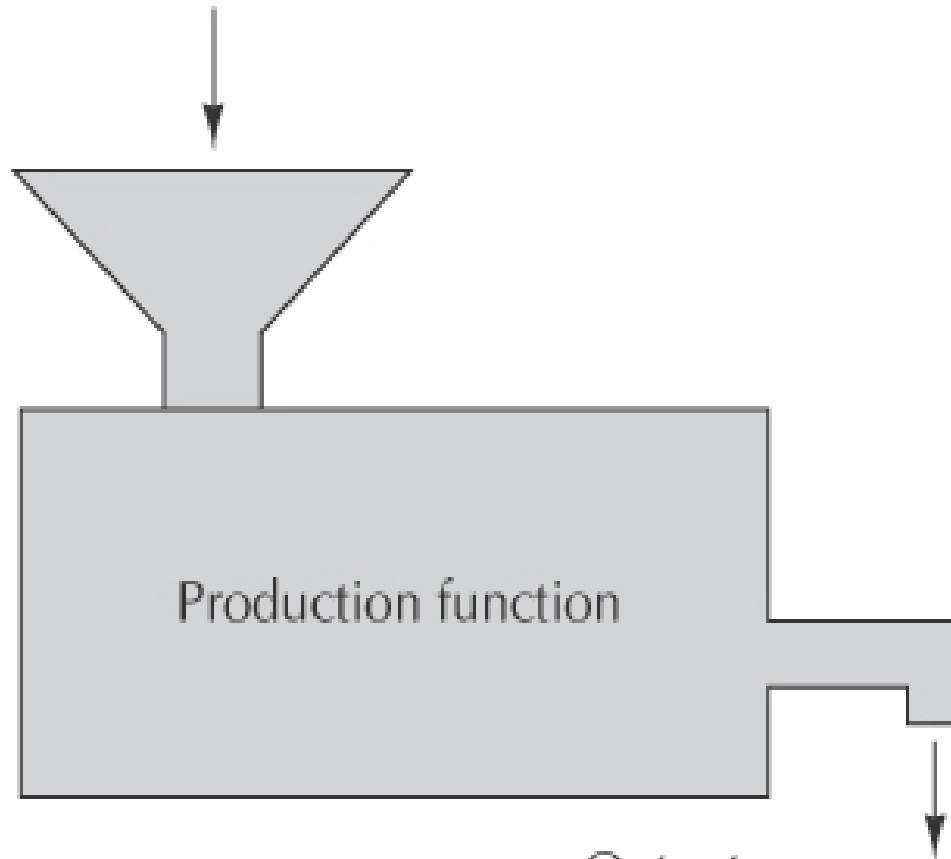
- **Production** refers to the process of transferring inputs into outputs.
- **Production function** is a technical relationship between inputs and outputs.
- Mathematically,

$$Q = f(X_1, X_2, X_3, \dots, X_n)$$

- $X_1, X_2, X_3, \dots, X_n$ are different types of inputs.
- More specifically,

$$Q = f(K, L)$$

Inputs
(land, labor, capital, and so forth)



Outputs
(cars, polio vaccine,
home-cooked meals,
FM radio broadcasts, and so forth)

Classification of inputs

- **Fixed inputs** - quantity remains constant(fixed) for a given period of time.
- **Variable inputs** –inputs that can change in quantity depending on the level of output being produced.

Production period

- **Short run** : refers to a period of time in which the quantity of at least one input remains fixed while others are variable
- Put it differently, one or more factors of production cannot be changed

Production period

- In this case, the firm can alter its level of output by increasing or decreasing the use of variable inputs.
- **Long run** : period of time in which all inputs used are variable.
- Here the period is long enough to allow changes in the level of all inputs
- i.e all inputs are variable and there is no fixed input .

3.2.The SR Production Function: one variable input

The contribution that variable input makes to the production process can be described in terms of the total, average and marginal product.

- **Total product (TP)**- total amount of output produced at different levels of inputs
- **Average product(AP)**: Average product of an input is the level of output that each unit of input produces, on the average **APL = TP/L**
- **The marginal product (MP)** - change in output as an input is changed by one unit.

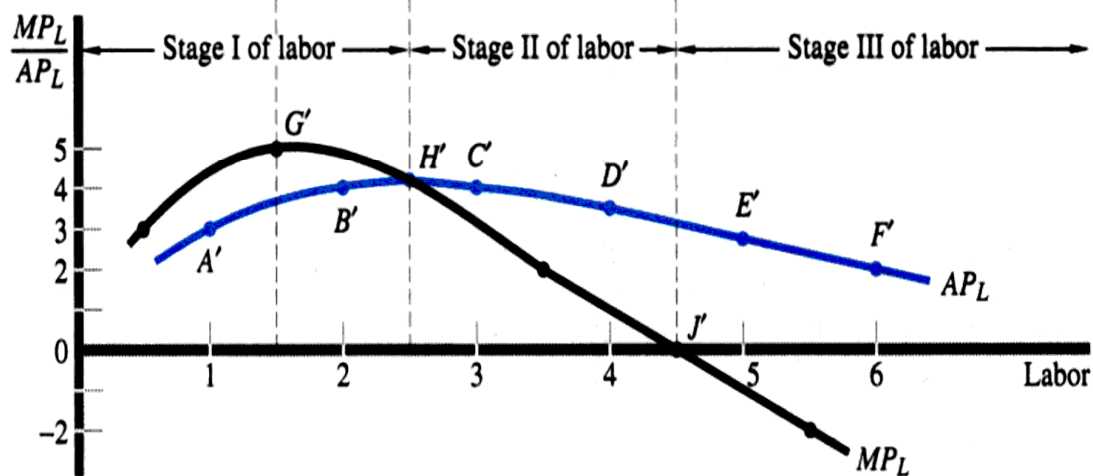
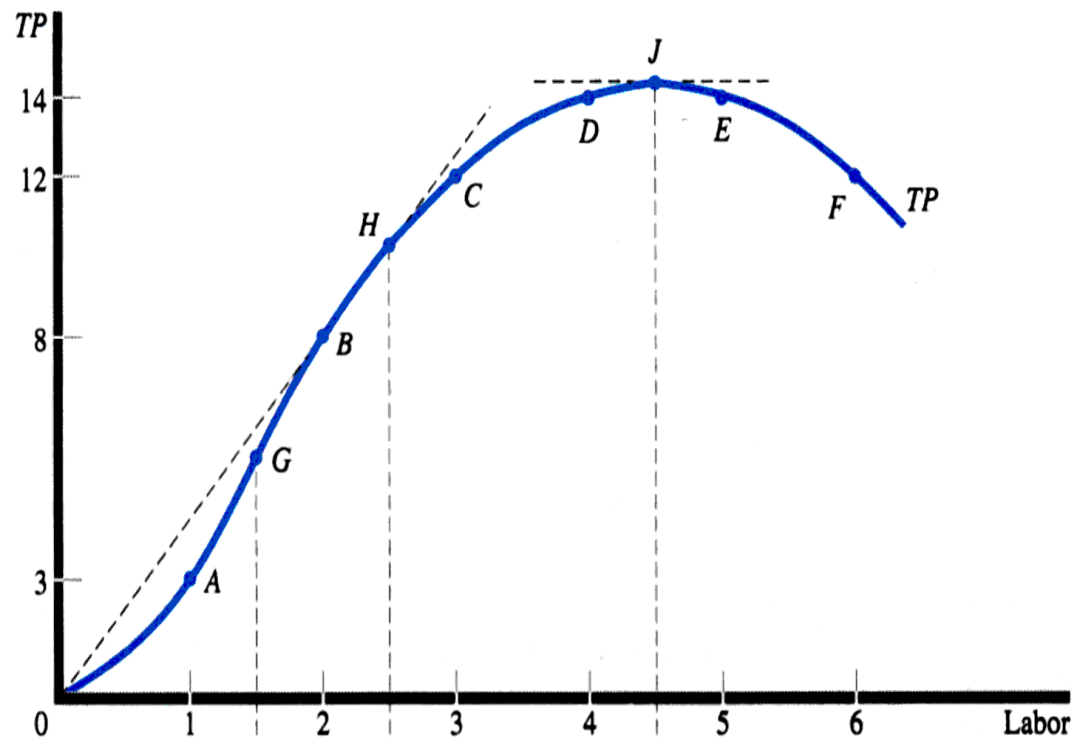
$$MP_L = \frac{\Delta TP}{\Delta L} \quad \text{or}$$

$$MP_L = \frac{dQ}{dL}$$

Cont...

the extra output brought about by the extra employment of a variable input. In terms of labor, we may ask “how much has the last laborer added to total product?” These issues are explained by the marginal concept.

L	TP_L	AP_L	MP_L
0	0	-	-
1	2	2	2
2	6	3	4
3	12	4	6
4	20	5	8
5	26	5.2	6
6	30	5	4
7	32	4.6	2
8	32	4	0
9	30	3.3	-2
10	26	2.6	-4



Relationship between MP_L & TP :

- When MPL is $\uparrow$ ing, TP is also $\uparrow$ ing at an $\uparrow$ ing rate
- When MPL is $\downarrow$ ing until it reaches zero, TP is $\uparrow$ ing but at a $\downarrow$ ing rate
- When $MPL = 0$, TP is maximum (constant level)
- When MPL is negative, TP is declining

Relationship between MP_L & AP_L :

- If $MP > AP$, AP is $\uparrow$ ing
- If $MP = AP$, AP is either at a maximum or is constant (H')
- If $MP < AP$, AP is $\downarrow$ ing

Numerical Illustration

- Suppose that the short-run production function for cut-flower by a certain Ethiopian firm is given by:

$$Q = 4KL - 0.6K^2 - 0.1L^2$$

where Q - represents the annual quantity of cut-flower produced.

K - annual capital input; suppose $K=5$.

L - annual labor input.

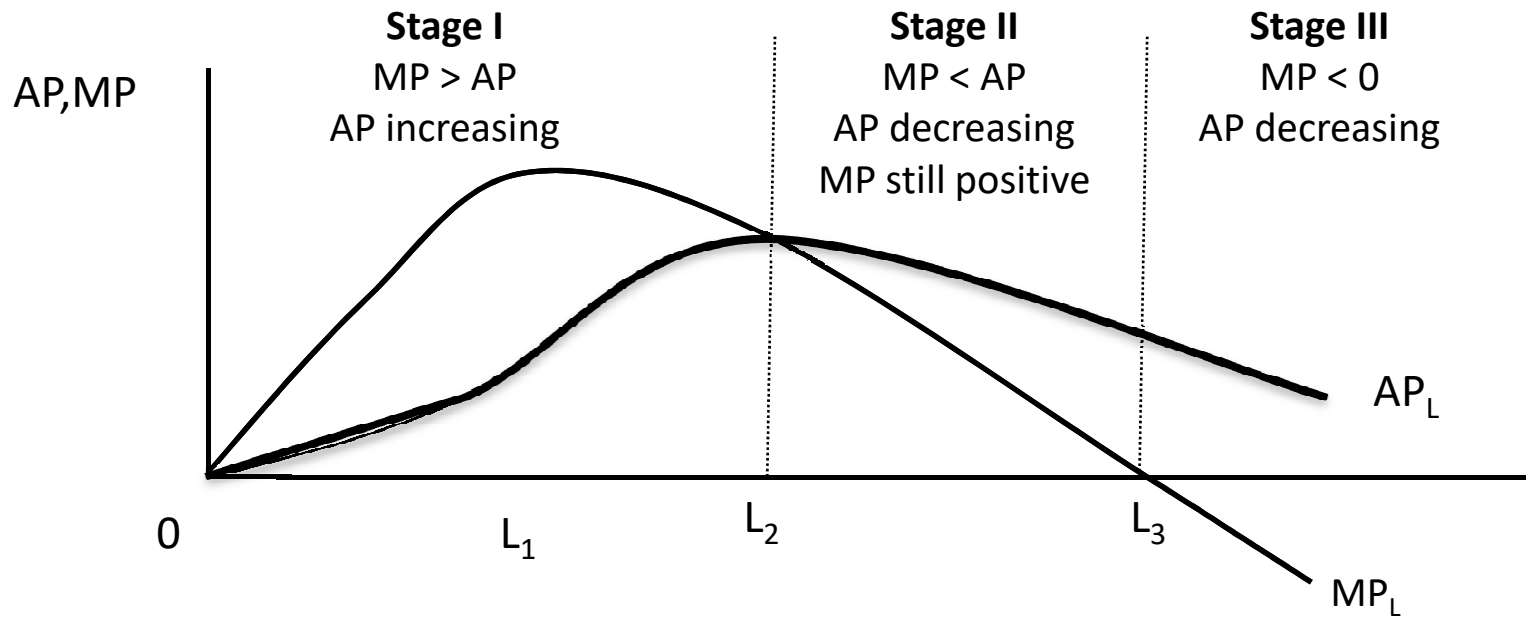
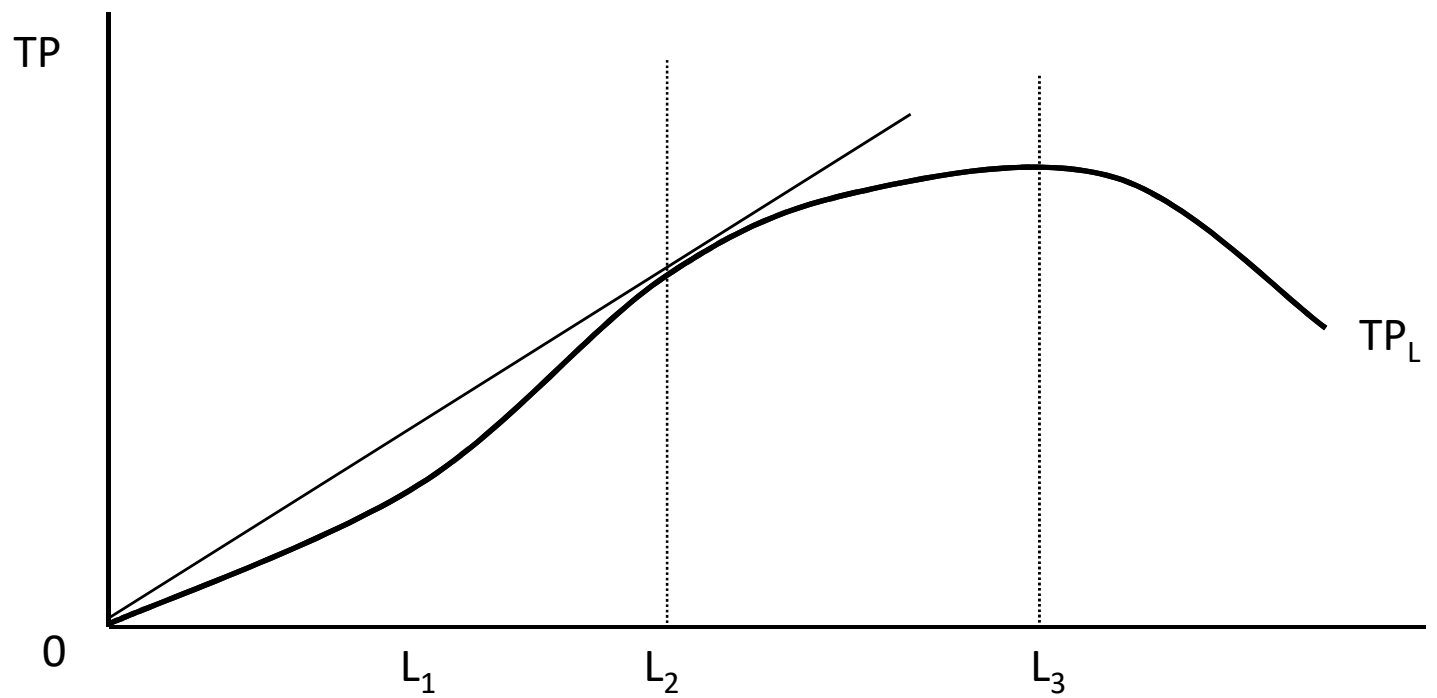
- a) Determine the average product of labor (AP_L) function.
- b) At what level of labor does the total output of cut-flower reach the maximum?
- c) What will be the maximum achievable amount of cut-flower production?

Law of Diminishing Marginal Returns

- The LDMP is the major factor behind the relationship between TP, MP, and AP.
- When additional units of a variable input are added to fixed inputs, after a certain point, the marginal product of the variable input declines.
- The Law states that if more and more units of a variable input are applied to a fixed input, the total output may initially increase at an increasing rate, but beyond a certain level of output, it increases at a diminishing rate. The law is sometimes called the **law of variable proportions**.

Stages of Production

- Stage I – Increasing return
- Stage 2– Diminishing return(**Economic region**)
- Stage 3 – Negative return
- **Increasing return**
- Marginal product of the variable input is higher than its AP ($MP > AP$), AP is increasing
- Goes from the origin (where $TP=0$) to the point where the AP is max ($APL=MPL$).



- **Stage II: Diminishing Returns**

- Marginal product of the variable input is falls below its AP ($MP < AP$), AP is $\downarrow$ ing $MP > 0$, and
- $MP > 0$ ranges from maximum AP_L to zero MP_L or it begins when AP_L starts to declining & ends when MP becomes zeros or reaches its maximum level.

- **Stage III: Negative Returns**

- Marginal product of the variable input/labor/turns negative ($MP < 0$)
- Thus, a rational consumer will not stop in stage I but will expand to stage II. Similarly a rational producer never prefers stage III since the return is negative..



Microeconomics

Theory of Cost

4.1. Basic concepts

- Cost is the monetary value of inputs used in production of an item.

- Types of cost of production

A. Social cost: is the cost of producing an item to the society.

This cost is realized due to the fact that most resources used for production purpose are scarce and some production process, by their nature, emit dangerous chemicals, bad smell, etc to surrounding society.

A. Private cost: is the cost of producing an item to the individual producer.

Private cost of production can be measured in two ways:

- **Economic cost** - Explicit cost plus Implicit cost
- **Accounting Cost** - Explicit cost of production

Costs as Opportunity Costs

- A firm's *cost of production* includes all the opportunity costs of making its output of goods and services.
- Explicit and Implicit Costs
 - A firm's cost of production include explicit costs and implicit costs.
 - *Explicit* costs are input costs that require a direct outlay of money by the firm.
 - *Implicit* costs are input costs that do not require an outlay of money by the firm.

Economic Profit versus Accounting Profit

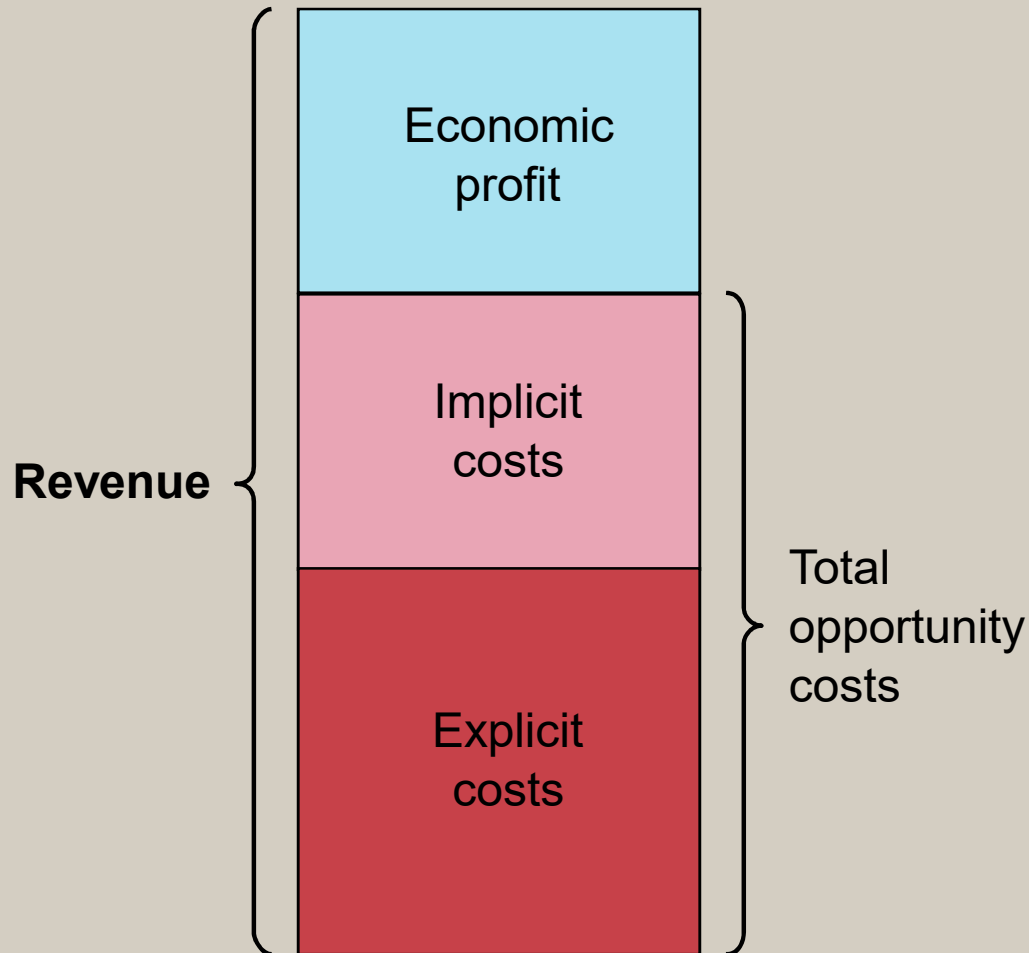
- Economists measure a firm's *economic profit* as total revenue minus total cost, including both explicit and implicit costs.
- Accountants measure the *accounting profit* as the firm's total revenue minus only the firm's explicit costs.

Economic Profit versus Accounting Profit

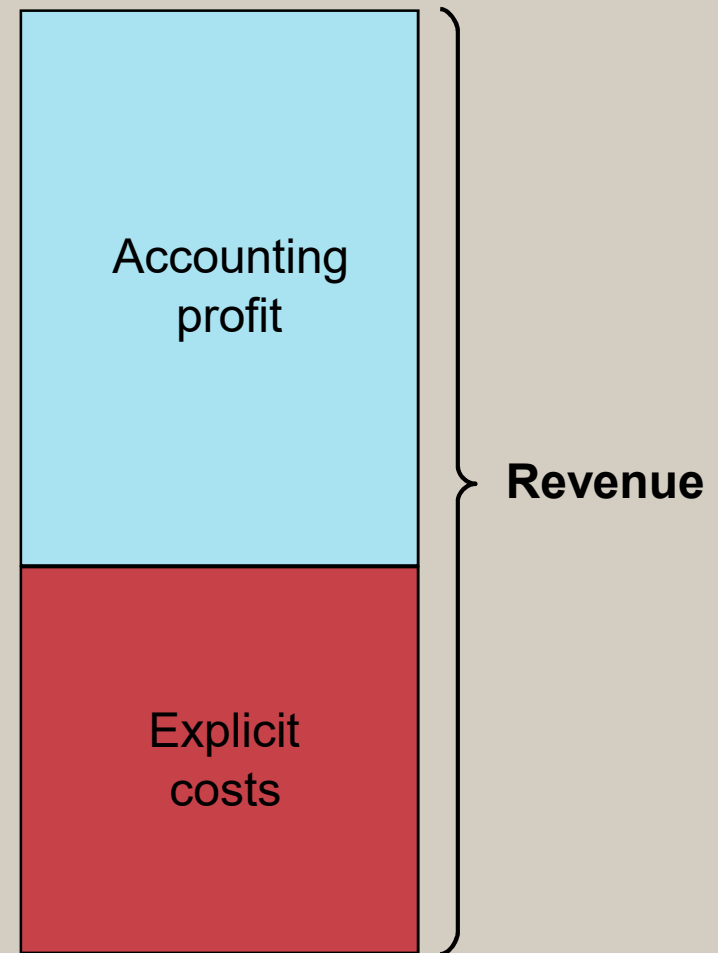
- When total revenue exceeds both explicit and implicit costs, the firm earns economic profit.
 - Economic profit is smaller than accounting profit.

Figure 1 Economic versus Accountants

How an Economist Views a Firm



How an Accountant Views a Firm



4.2. The SR Cost Function

- Cost function shows the algebraic relation between the cost of production and various factors which determine it.
- Among others, the cost of production depends on:
 - the level of output produced,
 - technology of production,
 - prices of factors, etc.

$$C = f(x, t, p_i)$$

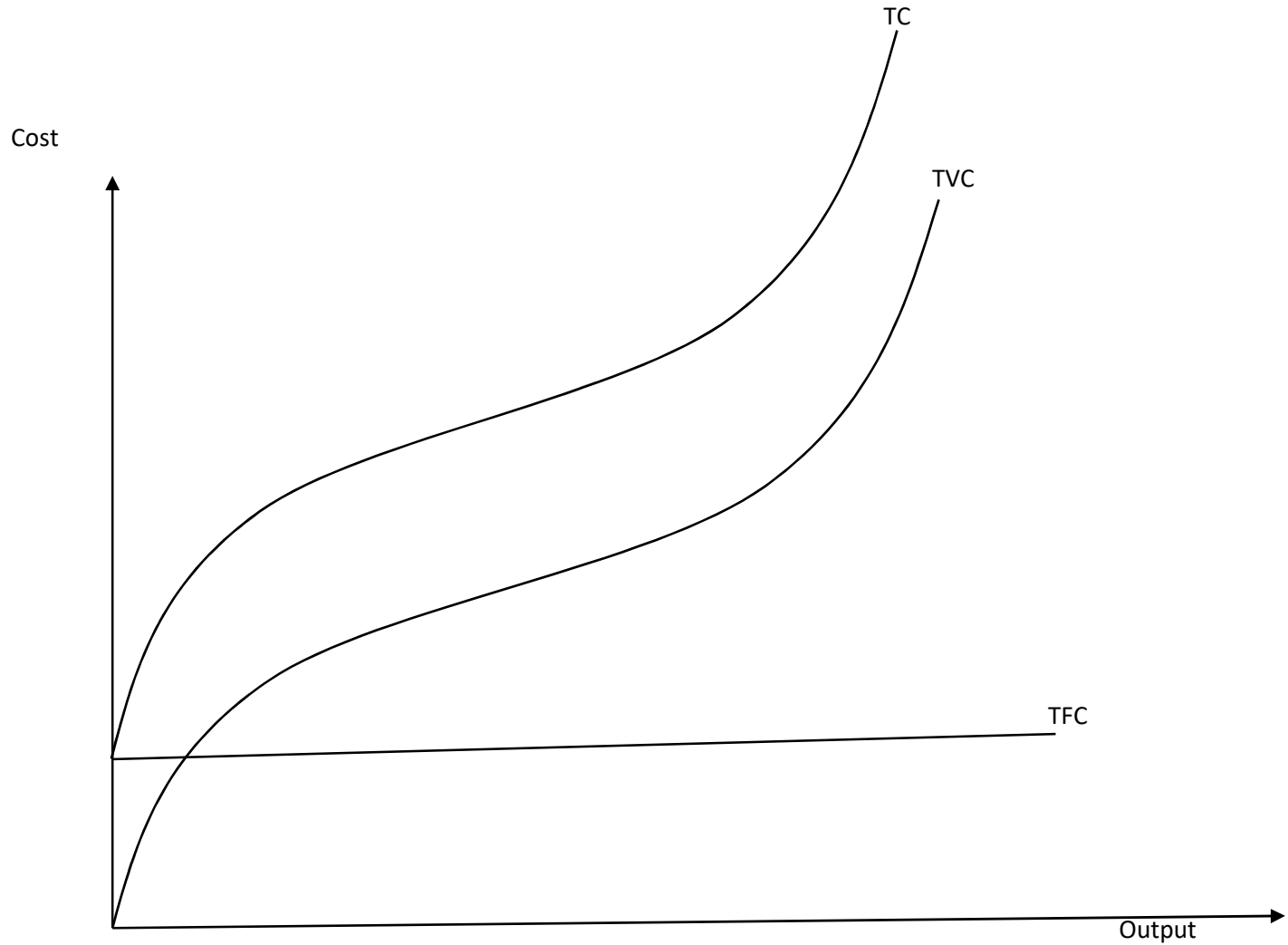
Short run costs of the traditional theory

- In the traditional theory of the firm, total costs are split into two groups: total fixed costs and total variable costs:

$$TC = TFC + TVC$$

- **Total fixed cost (TFC)** - a cost which doesn't vary with the level of output.
- **Total variable cost (TVC)** - include all costs which directly vary with the level of output.

Graphical presentation



Average costs

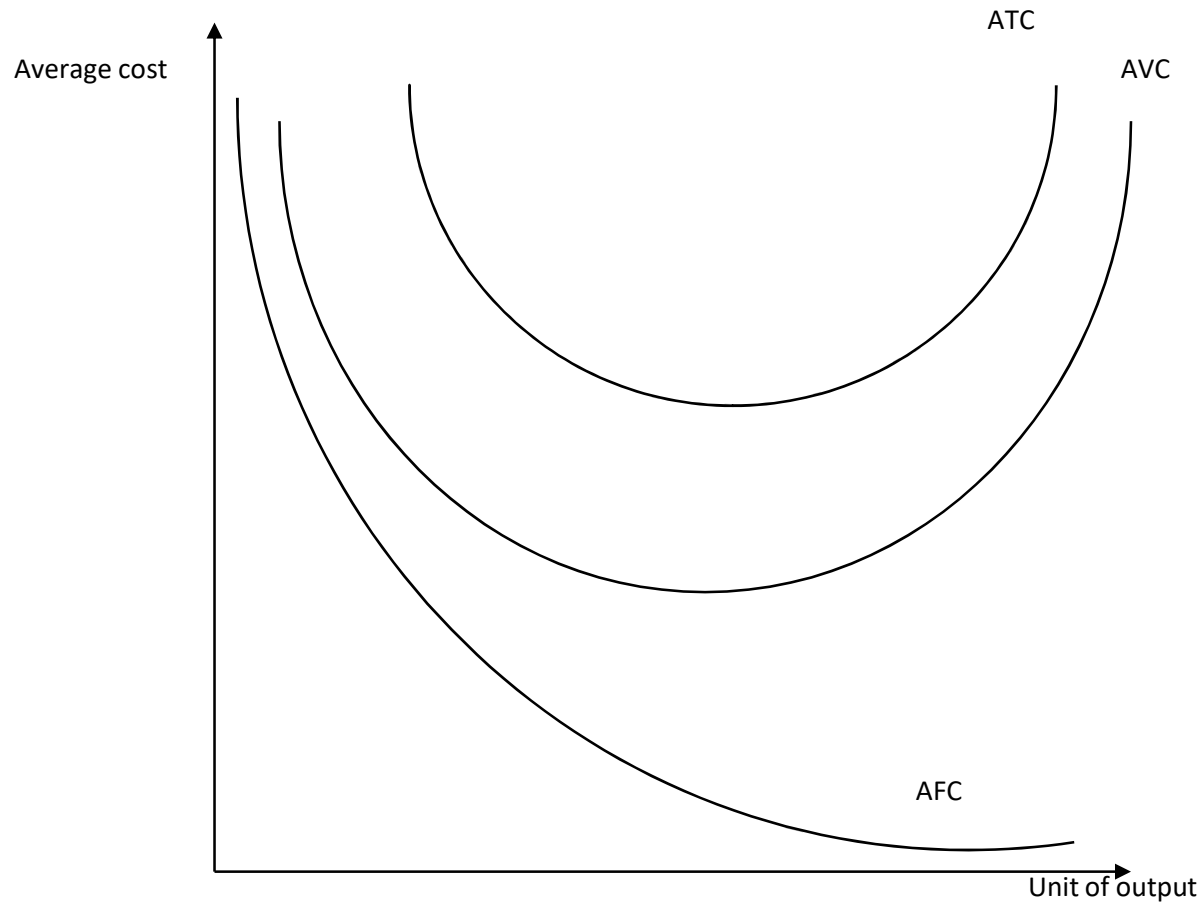
- Average costs are important as we may make comparisons with product price, which is always stated on per unit basis.
- **Average fixed costs (AFC)** – $AFC = \frac{TFC}{Q}$
- **Average variable cost (AVC)**

$$AVC = \frac{TVC}{Q}$$

- **Average total cost (ATC)**

$$ATC = \frac{TC}{Q} \quad \Rightarrow \quad \frac{TC}{Q} = \frac{TVC}{Q} + \frac{TFC}{Q}$$

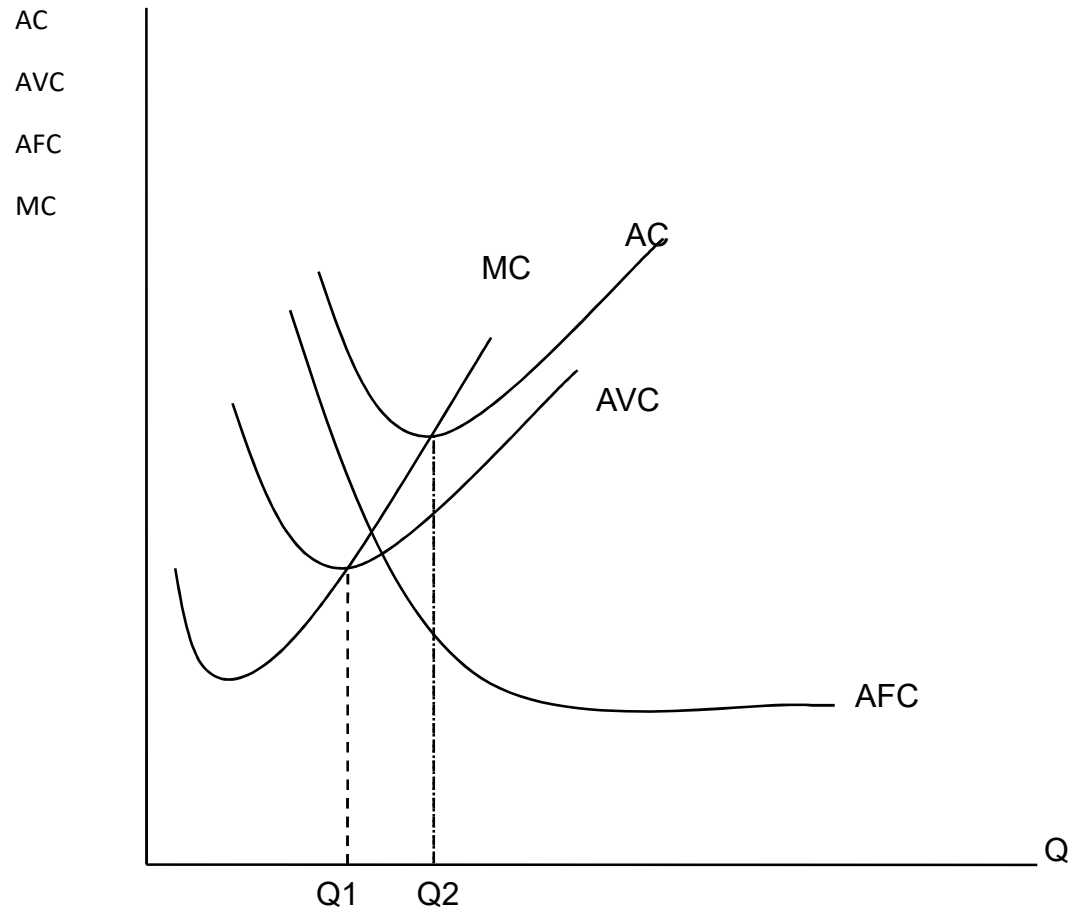
Graphical presentation



The relationship between AVC, ATC and MC

- MC is the extra cost of producing one more unit of output.
- The marginal cost curve intersects both the AVC and ATC curves at their minimum points.

Graphical presentation



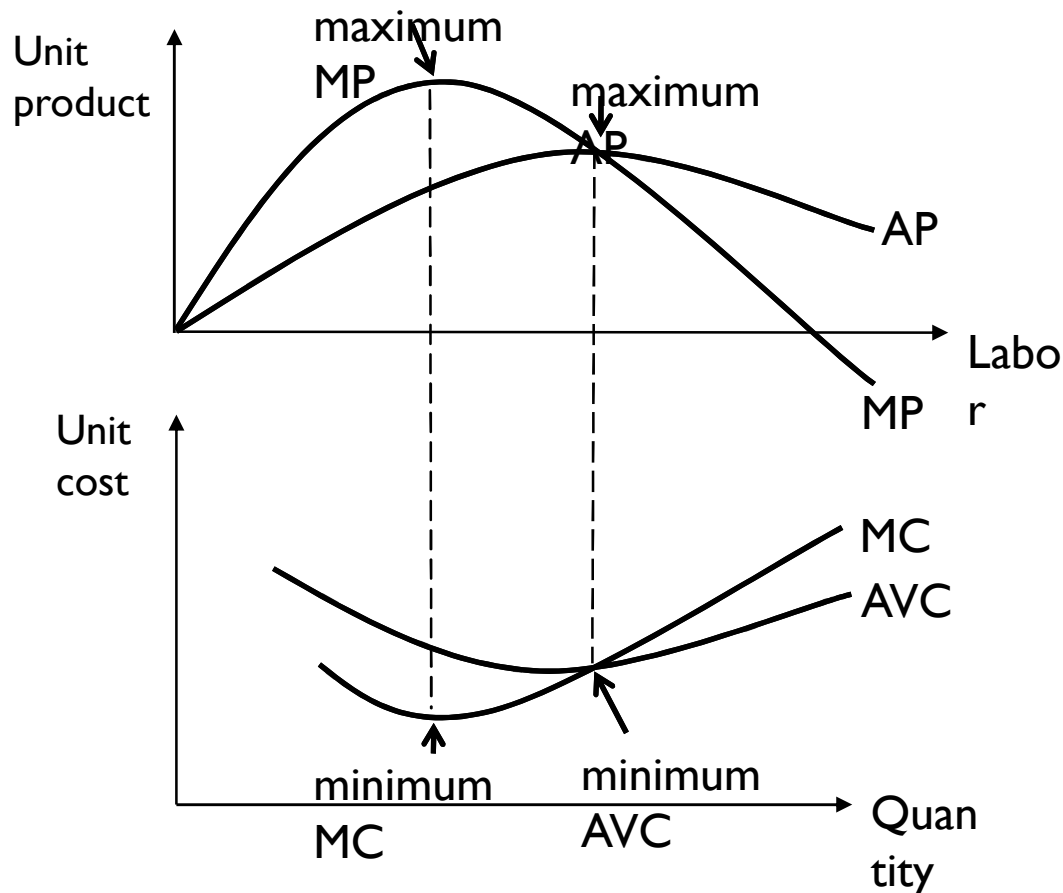
- The above graphical presentation can also be shown by using calculus.
 - Suppose the $TC = f(Q)$, then
 - Slope $AC = \frac{1}{Q}(MC - AC)$
 - How?
- Based on this the following relationships can be drawn:
 - when $MC < AC$, the slope of AC is negative
 - When $MC > AC$, the slope of AC is positive
 - When $MC = AC$, the slope of AC is zero

Numerical illustrations

- Suppose the short – run cost function of a firm is given by: $C=2Q^3 - 2Q^2 + Q + 10$,
Find:
 - The expressions for TFC & TVC
 - The expressions for AFC, AVC & AC and MC
 - The minimum values of MC and AVC.

Relationship between cost and production

- MC and AVC curves are mirror images of the MP and AP curves respectively.



$$AVC = w (L/Q)$$

$$\Rightarrow MC = \frac{w}{MP_L}$$

How?